Time Is of the Essence

Experimental Evidence on the Effects of Job Application Timing

Sai Zhang

University of Southern California

October 22, 2026

Timing in the Labor Market

- **Matching unfolds over time:** applications, evaluation, interviews, and offers

Timing in the Labor Market

- **Matching unfolds over time:** applications, evaluation, interviews, and offers
- **Job-seeker side:** a growing literature studies when to search, apply and accept offers

Timing in the Labor Market

- **Matching unfolds over time:** applications, evaluation, interviews, and offers
- **Job-seeker side:** a growing literature studies when to search, apply and accept offers
- **Firm side:** when to act on an evolving application pool?

Timing in the Labor Market

- **Matching unfolds over time:** applications, evaluation, interviews, and offers
- **Job-seeker side:** a growing literature studies when to search, apply and accept offers
- **Firm side:** when to act on an evolving application pool?
- **This paper:** how firms' timing decisions shape who gets considered

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications **receive more contact**?

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications **receive more contact**?

  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?

  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day

- What **drives** this early application advantage?

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?

  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day

- What **drives** this early application advantage?

  Firms **start contacting early**,
  triggered by the **best-so-far** application passing a threshold

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?

  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day

- What drives this early application advantage?

  Firms **start contacting early**,
  triggered by the **best-so-far** application passing a threshold

- How does it vary with **labor-market conditions**?

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?

  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day

- What drives this early application advantage?

  Firms **start contacting early**,
  triggered by the **best-so-far** application passing a threshold

- How does it vary with **labor-market conditions**?

  The timing gradient is **steeper** in **tighter** labor markets

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?
  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day
- What drives this early application advantage?
  Firms **start contacting early**,
  triggered by the **best-so-far** application passing a threshold
- How does it vary with labor-market conditions?
  The timing gradient is **steeper** in **tighter** labor markets
- What is the **cost of speed** for firms?

In This Paper

Repurposing the resume correspondence experiment of Kline et al. (2022), I ask:

- Do earlier applications receive more contact?
  1.2 p.p. (*57% of the racial contact gap*); **0.4 p.p.** per day
- What drives this early application advantage?
  Firms **start contacting early**,
  triggered by the **best-so-far** application passing a threshold
- How does it vary with labor-market conditions?
  The timing gradient is **steeper** in **tighter** labor markets
- What is the **cost of speed** for firms?
  A stopping model quantifies the tradeoff: **6.6%** relative loss in match quality

Literature Contribution

- **Timing in the labor market**

  Bandiera et al. (2025), Cortés et al. (2023), and Davis and de la Parra (2024)

  - **This paper: firm-side timing of contact initiation**

Literature Contribution

- **Timing in the labor market**

  Bandiera et al. (2025), Cortés et al. (2023), and Davis and de la Parra (2024)

  - **This paper: firm-side timing of contact initiation**

- **Employer screening and recruiting strategy**

  Bartoš et al. (2016), Hoffman et al. (2018), and Radbruch and Schiprowski (2025)

  - **This paper: a threshold-based screening strategy for when firms act**

Literature Contribution

- **Timing in the labor market**
  Bandiera et al. (2025), Cortés et al. (2023), and Davis and de la Parra (2024)
  - **This paper: firm-side timing of contact initiation**
- **Employer screening and recruiting strategy**
  Bartoš et al. (2016), Hoffman et al. (2018), and Radbruch and Schiprowski (2025)
  - **This paper: a threshold-based screening strategy for when firms act**
- **Recruiting and matching efficiency**
  Birinci et al. (2025), Carrillo-Tudela et al. (2023), Wolthoff (2018), and Wright et al. (2021)
  - **This paper: a within-vacancy timing margin of match formation**

Literature Contribution

- **Timing in the labor market**
  Bandiera et al. (2025), Cortés et al. (2023), and Davis and de la Parra (2024)
  - **This paper: firm-side timing of contact initiation**
- **Employer screening and recruiting strategy**
  Bartoš et al. (2016), Hoffman et al. (2018), and Radbruch and Schiprowski (2025)
  - **This paper: a threshold-based screening strategy for when firms act**
- **Recruiting and matching efficiency**
  Birinci et al. (2025), Carrillo-Tudela et al. (2023), Wolthoff (2018), and Wright et al. (2021)
  - **This paper: a within-vacancy timing margin of match formation**
- **Audit-study design**
  Gaddis (2018), Kessler et al. (2019), and Neumark (2018)
  - **This paper: multiple applications as tracers**

Early Application Advantage

Experiment Design (Kline et al., 2022)



vacancy open

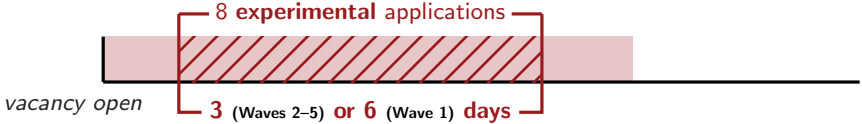
Experiment Design (Kline et al., 2022)



vacancy open

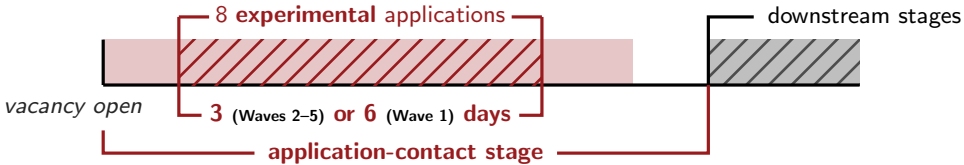
- **11,114** entry-level vacancies at 108 large U.S. firms

Experiment Design (Kline et al., 2022)



- **11,114** entry-level vacancies at 108 large U.S. firms
- **8 randomized applications** in 4 pairs over 3 (Waves 2–5) or 6 days (Wave 1)

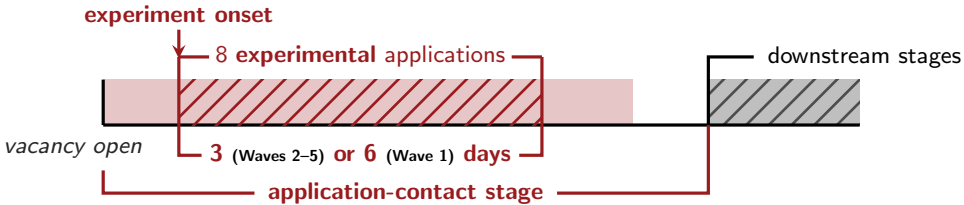
Experiment Design (Kline et al., 2022)



- **11,114** entry-level vacancies at 108 large U.S. firms
- **8 randomized applications** in 4 pairs over 3 (Waves 2–5) or 6 days (Wave 1)
- **outcome**: employer contact within 30 days (∼24%)

▸ design details

Experiment Design (Kline et al., 2022)



- **11,114** entry-level vacancies at 108 large U.S. firms
- **8 randomized applications** in 4 pairs over 3 (Waves 2–5) or 6 days (Wave 1)
- **outcome**: employer contact within 30 days (∼24%)
- **Market conditions** measured at the first experimental submission

▸ design details
▸ covariates

Submission Order and Sample Completeness

Order	Full Sample	*Incomplete*
1	**11,114**	1,285
2	11,031	1,202
3	10,682	853
4	10,606	777
5	10,278	449
6	10,204	375
7	9,899	70
8	**9,829**	0
N applications	83,643	5,011
N vacancies	11,114	*1,285*
N firms	108	39

balance

Submission Order and Sample Completeness

Order	Full Sample	*Incomplete*	Wave 1	Waves 2–5
1	**11,114**	1,285	2,063	9,051
2	11,031	1,202	2,040	8,991
3	10,682	853	1,883	8,799
4	10,606	777	1,879	8,727
5	10,278	449	1,748	8,530
6	10,204	375	1,734	8,470
7	9,899	70	1,604	8,295
8	**9,829**	0	1,590	8,239
N applications	83,643	5,011	14,541	69,102
N vacancies	11,114	*1,285*	2,063	9,051
N firms	108	39	85	105

▸ balance

Measuring Application Timing: Early Indicator

Define the first 4 submissions (out of 8) to a job as **early**:

$$early_i = \mathbf{1}(\text{application } i \text{ among the first 4 applications})$$

41,112 submissions (49.15%) are classified as early.

Measuring Application Timing: Early Indicator

$$early_i = \mathbf{1}\,(\text{application } i \text{ among the first 4 applications})$$

Estimate

$$Y_i = \alpha_0 + \beta \cdot early_i + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

where

- $\mathbf{X}_i$ includes the randomized resume characteristics, $\eta_{j(i)}$ is a vacancy fixed effect.
- Standard errors are clustered at the vacancy level (Abadie et al., 2023).

Measuring Application Timing: Early Indicator

$$early_i = \mathbf{1}\,(\text{application } i \text{ among the first 4 applications})$$

Estimate

$$Y_i = \alpha_0 + \beta \cdot early_i + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

$early_i$	$Black_i$	$Female_i$
0.0116***	**-0.0205*****	**-0.0021**
(0.0024)	(0.0018)	(0.0022)

Full sample: $N = 83,643$

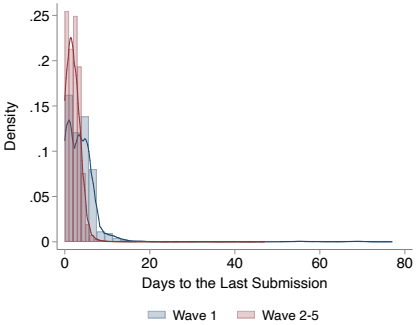
▸ robustness ▸ by order

Measuring Application Timing: Days to Last Submission

Let $D2L_i$ be the number of days between application i and the **final experimental submission** to the same vacancy,

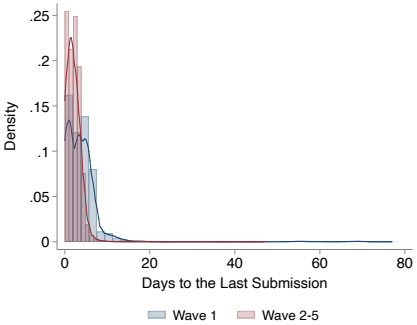
Measuring Application Timing: Days to Last Submission

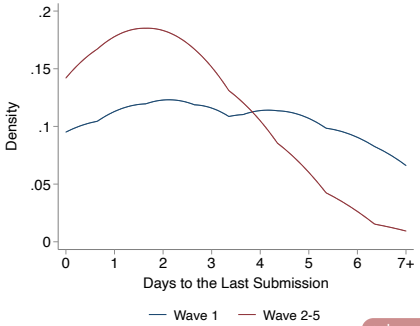
Let $D2L_i$ be the number of days between application i and the **final experimental submission** to the same vacancy,



Measuring Application Timing: Days to Last Submission

Let $D2L_i$ be the number of days between application i and the **final experimental submission** to the same vacancy, define $D2L_i^t = \min\{D2L_i, 7\}$.





▸ by order

Early Application Advantage: Days to Last Submission

Estimate

$$Y_i = \alpha_0 + \delta \cdot D2L_i^t + \boldsymbol{\gamma}' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1) Full Sample
$D2L_i^t$	**0.0040*****
	(0.0008)
N	83,643

▸ robustness

Early Application Advantage: Days to Last Submission

Estimate

$$Y_i = \alpha_0 + \delta \cdot D2L_i^t + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1) Full Sample	(2) Wave 1	(3) Waves 2–5
$D2L_i^t$	**0.0040*****	0.0036***	0.0043***
	(0.0008)	(0.0013)	(0.0010)
N	83,643	14,541	69,102

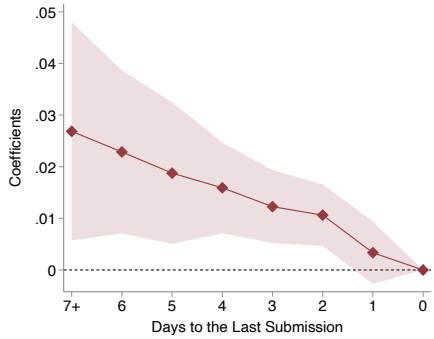
▸ robustness

Early Application Advantage: Days to Last Submission

$$Y_i = \alpha + \sum_{d=1}^{7} \delta_d \mathbf{1}\{D2L_i^t = d\} + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

Early Application Advantage: Days to Last Submission

$$Y_i = \alpha + \sum_{d=1}^{7} \delta_d \mathbf{1}\{D2L_i^t = d\} + \boldsymbol{\gamma}' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$



▸ robustness ▸ heterogeneity

Mechanism: Contact Initiation

A Firm-Side Mechanism



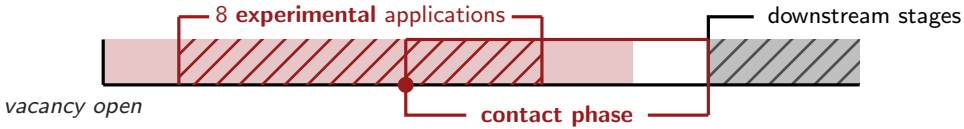
- **So far:** Earlier experimental applications are more likely to be contacted

A Firm-Side Mechanism



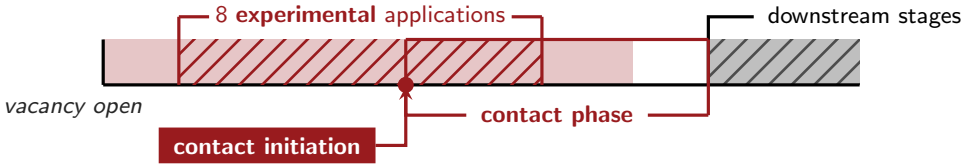
- **So far:** Earlier experimental applications are more likely to be contacted
- **Possible families of explanations**: **direct-timing** versus **pool-based**

A Firm-Side Mechanism



- **So far:** Earlier experimental applications are more likely to be contacted
- **Possible families of explanations**: **direct-timing** versus **pool-based**
- **Hypothesis:** Applications arriving in the **contact phase** face a different recruiting environment

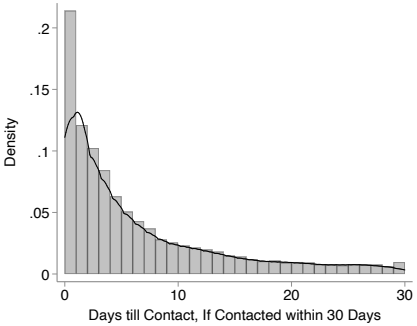
A Firm-Side Mechanism



- **So far:** Earlier experimental applications are more likely to be contacted
- **Possible families of explanations**: **direct-timing** versus **pool-based**
- **Hypothesis:** Applications arriving in the **contact phase** face a different recruiting environment
- **Key state change:** the timing of **contact initiation**

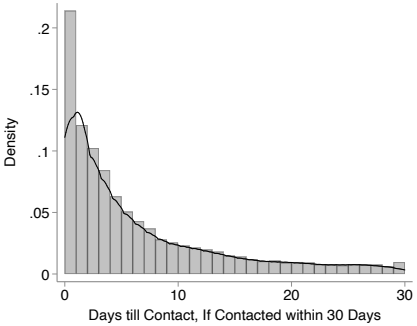
Recruiters Initiate Contact Early

Applications: Contacts happen fast

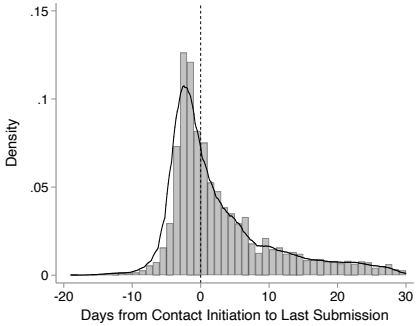


Recruiters Initiate Contact Early

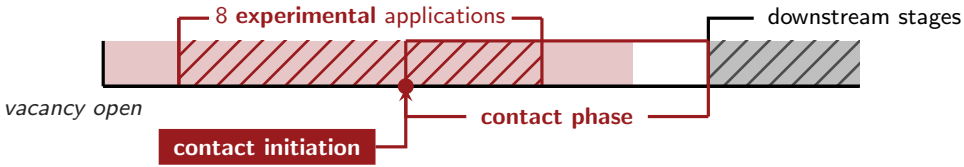
Applications: Contacts happen fast



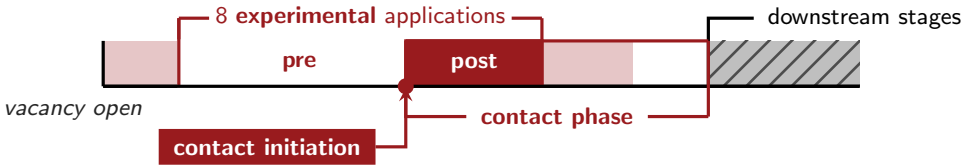
Vacancies: Contact initiation is early



Contact Initiation: Changes in the Recruiting Environment



Contact Initiation: Changes in the Recruiting Environment

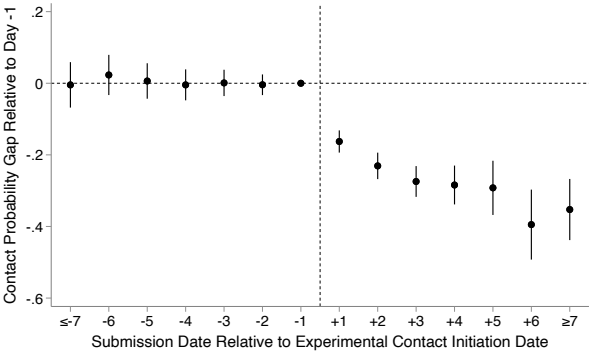


Estimate:

$$Y_i = \sum_{k \in \mathcal{K}, k \neq -1} \beta_k \cdot \mathbf{1}\left(e_{j(i)} = k\right) + \boldsymbol{\gamma}' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i,$$

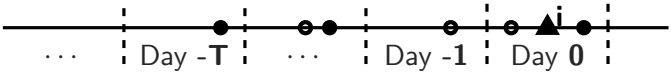
where $e_{j(i)} = \text{submission date}_i - \text{contact initiation date}_{j(i)}$, $\mathbf{X}_i$ includes randomized resume controls, submission-order FEs, and submission-day-of-week FEs.

Contact Initiation: Changes in the Recruiting Environment

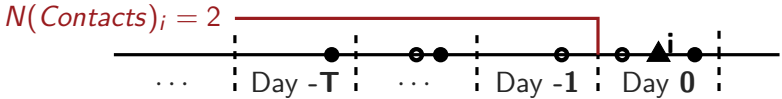


Contact probability drops sharply **after contact initiation**

Constructing the Post-Initiation Indicator

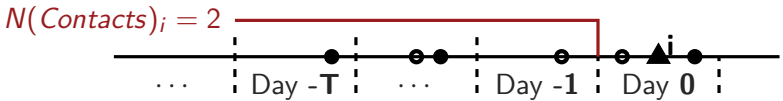


Constructing the Post-Initiation Indicator



If application i arrives at least one day after contact initiation, $N(Contacts)_i > 0$.

Constructing the Post-Initiation Indicator



If application i arrives at least one day after contact initiation, $N(Contacts)_i > 0$.

	$N(Contacts)_i = 0$	$N(Contacts)_i > 0$	Diff.
	.2641***	.0974***	**-.1667****
	(.0034)	(.0125)	(.0127)
N	70,293	8,339	78,632

Contact probability decreases by **63%** after contact initiation.

▸ $N(Contacts)_i^{alt}$

Contact Initiation Absorbs the Timing Advantage

$$Y_i = \alpha_0 + \beta \cdot timing_i + \iota \cdot \mathbf{1}\left(N(Contacts)_i > 0\right) + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1)	(2)	(3)	(4)
$early_i$	0.0112***	-0.0099***		
	(0.0024)	(0.0022)		
$D2L_i^t$			0.0038***	-0.0039***
			(0.0008)	(0.0007)
$\mathbf{1}(N(Contacts)_i > 0)$		-0.1733***		-0.1760***
		(0.0086)		(0.0087)
N		78,632		

▸ $N(Contacts)_i^{alt}$

Contact Initiation Absorbs the Timing Advantage

$$Y_i = \alpha_0 + \beta \cdot timing_i + \iota \cdot \mathbf{1}\left(N(Contacts)_i > 0\right) + \boldsymbol{\gamma}'\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

	(1)	(2)	(3)	(4)
$early_i$	0.0112***	-0.0099***		
	(0.0024)	(0.0022)		
$D2L_i^t$			0.0038***	-0.0039***
			(0.0008)	(0.0007)
$\mathbf{1}(N(Contacts)_i > 0)$		-0.1733***		-0.1760***
		(0.0086)		(0.0087)
N		78,632		

But contact initiation is **endogenous**.

▸ $N(Contacts)_i^{alt}$

Instrumenting Contact Initiation



- **Intuition:** the quality of applications before i is **randomized**

▸ model details ▸ $\hat{p}_{ij}$ validation

Instrumenting Contact Initiation



- **Intuition:** the quality of applications before i is **randomized**
- **Quality proxy:** train prediction models on the **first experimental application** to each vacancy

$$\widehat{p}_{ij} = \Pr(\text{contact}_{ij} = 1 \mid \mathbf{x}_{ij}, \mathbf{z}_j)$$

▸ model details ▸ $\hat{p}_{ij}$ validation

Instrumenting Contact Initiation



- **Intuition:** the quality of applications before i is **randomized**
- **Quality proxy:** train prediction models on the **first experimental application** to each vacancy

$$\widehat{p}_{ij} = \Pr(\text{contact}_{ij} = 1 \mid \mathbf{x}_{ij}, \mathbf{z}_j)$$

- **Instrument:** best out-of-fold predicted screening value **at least one day before** i

$$Q_i^{best} = \max_{k:day(k)<day(i)} \{\hat{p}_{kj}\} \quad \rightarrow \mathbf{1}\left(N(Contacts)_i > 0\right)$$

▸ model details ▸ $\hat{p}_{ij}$ validation

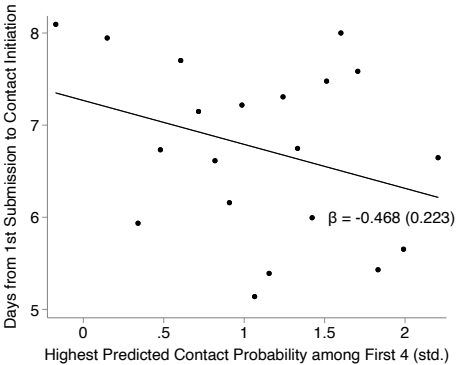
IV Estimation: Contact Initiation and Subsequent Contacts

$$Q_i^{best} \xrightarrow{\text{first stage}} \mathbf{1}\left(N(Contacts)_i > 0\right) \xrightarrow{\text{second stage}} \text{contact}$$

	All Vacancies		Contacted Vacancies	
	(1) IV-2SLS	(2) OLS	(3) IV-2SLS	(4) OLS
$\mathbf{1}\left(N(Contacts)_i > 0\right)$	**-1.2642***	-0.1204***	**-1.1077***	-0.1275***
	(0.7380)	(0.0206)	(0.5950)	(0.0217)
First Stage	0.0078**		0.0235**	
	(0.0039)		(0.0098)	
Kleibergen-Paap F	5.218		7.602	
Anderson-Rubin p	0.025		0.023	
N	39,300		16,142 (4,036 *vacancies*)	

▸ robustness

Best-So-Far Application as Contact Initiation Trigger



Higher Early Screening Value, **Earlier Contact Initiation**

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application $\rightarrow$ **contact initiation** $\rightarrow$ **lower subsequent contact**

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application → **contact initiation** → **lower subsequent contact**

The evidence also weighs against four alternatives:

- **Pool-quality learning**: Best-so-far predicts initiation, **mean-so-far does not**

▸ A

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application → **contact initiation** → **lower subsequent contact**

The evidence also weighs against four alternatives:

- **Pool-quality learning**: Best-so-far predicts initiation, **mean-so-far does not** ▸ A
- **Intrinsic value of earliness**: **The advantage disappears** for vacancies with contact initiation after all experimental submissions ▸ B

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application → contact initiation → lower subsequent contact

The evidence also weighs against four alternatives:

- **Pool-quality learning**: Best-so-far predicts initiation, **mean-so-far does not** ▸ A
- **Intrinsic value of earliness**: **The advantage disappears** for vacancies with contact initiation after all experimental submissions ▸ B
- **Full-pool comparison**: Screening value of later applications **does not predict earlier contacts** ▸ C

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application → contact initiation → lower subsequent contact

The evidence also weighs against four alternatives:

- **Pool-quality learning**: Best-so-far predicts initiation, **mean-so-far does not** ▸ A
- **Intrinsic value of earliness**: **The advantage disappears** for vacancies with contact initiation after all experimental submissions ▸ B
- **Full-pool comparison**: Screening value of later applications **does not predict earlier contacts** ▸ C
- **Pure batching**: Later submissions are generally **contacted later** ▸ D

The Evidence Points to Threshold-Based Screening

The evidence supports a threshold-based mechanism:

stronger best-so-far application $\rightarrow$ **contact initiation** $\rightarrow$ **lower subsequent contact**

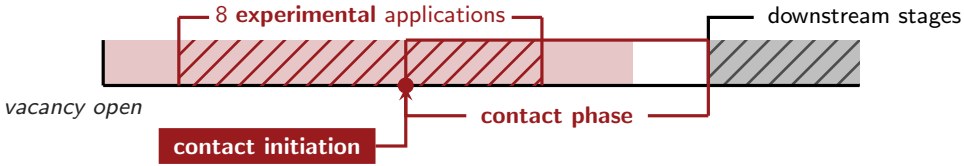
The evidence also weighs against four alternatives:

- **Pool-quality learning**: Best-so-far predicts initiation, **mean-so-far does not** ▸ A
- **Intrinsic value of earliness**: **The advantage disappears** for vacancies with contact initiation after all experimental submissions ▸ B
- **Full-pool comparison**: Screening value of later applications **does not predict earlier contacts** ▸ C
- **Pure batching**: Later submissions are generally **contacted later** ▸ D

The key firm choice: Initiate contact when the best so far **crosses a threshold**

A Model of Contact Initiation

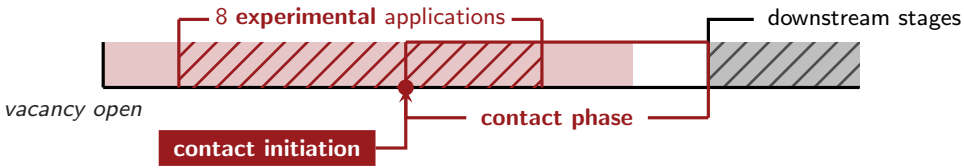
Application Timing in the Labor Market



So far:

- significant early application advantages
- driven by recruiters initiating contacts before the application pool is complete
- contact initiation triggered by best-so-far passing a threshold

Application Timing in the Labor Market

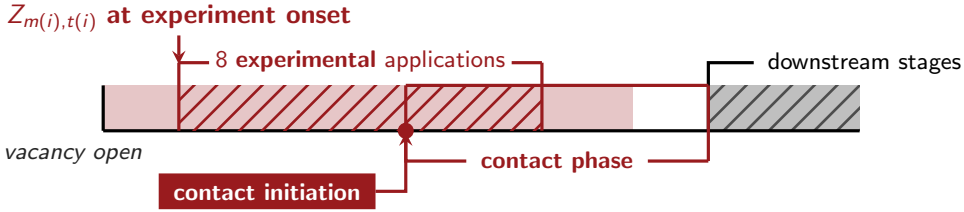


So far:

- significant early application advantages
- driven by recruiters initiating contacts before the application pool is complete
- contact initiation triggered by best-so-far passing a threshold

How does the timing of contact initiation vary with **labor-market conditions**?

Application Timing in the Labor Market



$$
\begin{aligned}
Y_i =& \alpha_0 + \delta D2F_i^t + \phi\left(D2F_i^t \times Z_{m(i),t(i)}\right) \\
&+ \phi^{\text{pre}}\left(D2F_i^t \times \bar{Z}_{m(i),t(i)-3:t(i)-1}\right) + \boldsymbol{\gamma}\mathbf{X}_i + \eta_{j(i)} + \varepsilon_i,
\end{aligned}
$$

$Z_{m(i),t(i)}$: conditions in local labor market $m(i)$ in the month of the **first submission** $t(i)$

Contact Timing by Labor Market Tightness

Dep. Var.	Contacted	Contact Initiation
$D2F_i^t$	-.0026***	-0.0135***
	(.0010)	(.0006)
	[.0010]	[.0008]
$D2F_i^t \times \log\left(V/U_{m(i),t(i)}\right)$	-0.0078	**-0.0291*****
	(.0057)	(.0040)
	[.0056]	[.0049]
N	78,632	78,632

▸ $D2F_i^t$ ▸ JOLTS

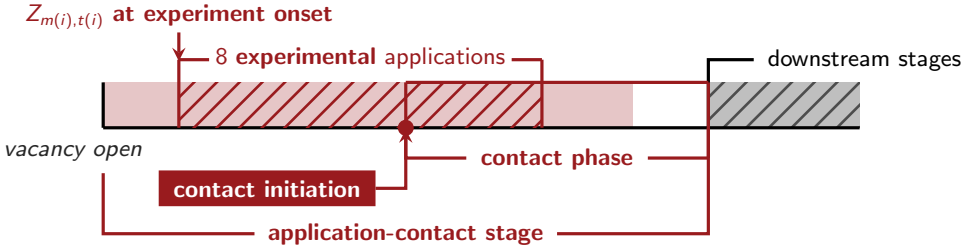
Contact Timing by Labor Market Tightness

Dep. Var.	Contacted	Contact Initiation
$D2F_i^t$	-.0026***	-0.0135***
	(.0010)	(.0006)
	[.0010]	[.0008]
$D2F_i^t \times \log\left(V/U_{m(i),t(i)}\right)$	-0.0078	**-0.0291*** **
	(.0057)	(.0040)
	[.0056]	[.0049]
N	78,632	78,632

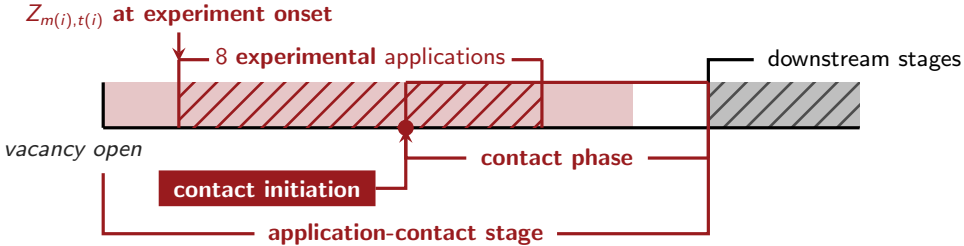
more vacancies to fill $\Rightarrow$ firms move faster $\Rightarrow$ early advantage $\uparrow$

▸ $D2F_i^t$ ▸ JOLTS

Modeling the Application-Contact Stage



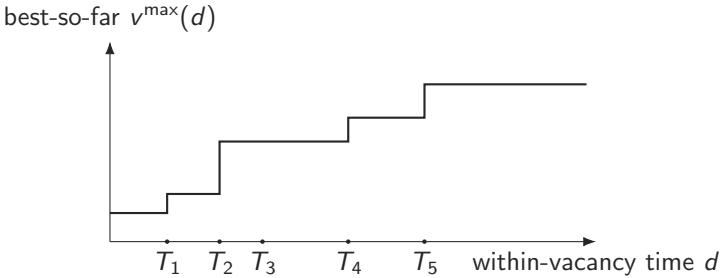
Modeling the Application-Contact Stage



Model: **a partial-equilibrium firm-side policy rule** that

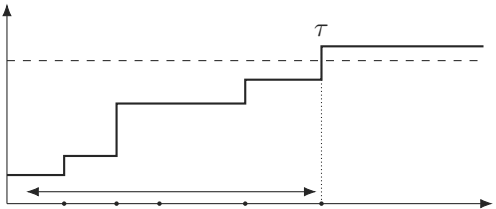
- quantifies the **recruiting-speed** versus **screening-value tradeoff** of firms
- accounts for **organic application inflows**
- separates two underlying economic forces: **arrival intensity** and **the cost of delay**

The Threshold Rule and Firms' Tradeoff



The Threshold Rule and Firms' Tradeoff

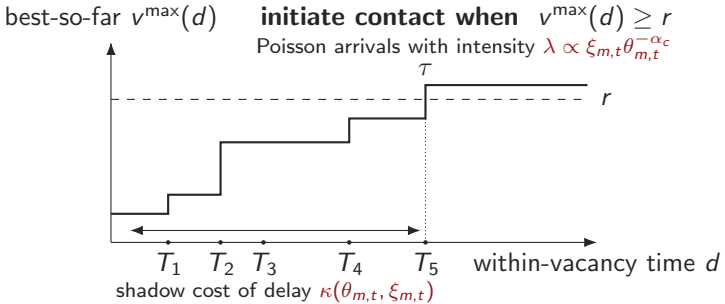
best-so-far $v^{\max}(d)$ **initiate contact when** $v^{\max}(d) \geq r$



The firm's optimization problem is

$$\max_{\tau} \mathbb{E}\Big[\underbrace{e^{-\delta\tau} v^{\max}(\tau)}_{\text{value of the best application}} - \underbrace{\int_0^{\tau} e^{-\delta s} \kappa ds}_{\text{cost of delay}} \Big]$$

The Threshold Rule and Firms' Tradeoff



The optimal threshold is characterized by

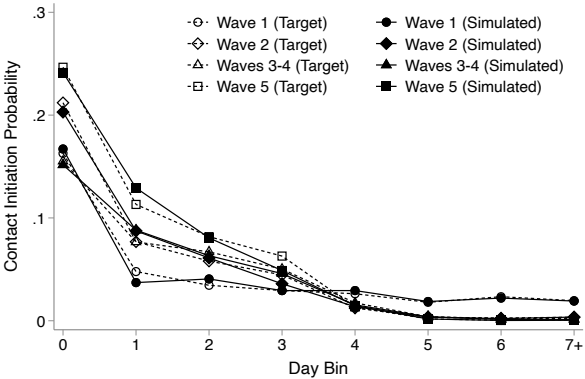
$$\underbrace{\delta r + \kappa}_{\text{cost of delay}} = \underbrace{\lambda \mathbb{E}\left[(v-r)^+\right]}_{\text{option value of another applicant}}, v \sim \mathcal{N}(0,1)$$

Identification and Estimation

- **Observed**: submission dates and experimental contact initiators
- **Mapping**: threshold crossing and survival from earlier initiation
- **Estimation**: Method of Simulated Moments
- **Two families of moments**: 33 moments, 9 or 12 parameters
  - Experimental initiator mass by day ($D2F_i^t = 0, \cdots, 6$), and residual *none* mass
  - tightness gradient of contact initiation
- Contact-initiation policy on a **normalized screening-value** scale

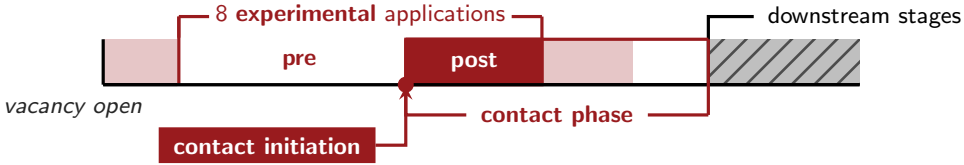
▸ targeted moments

Identification and Estimation

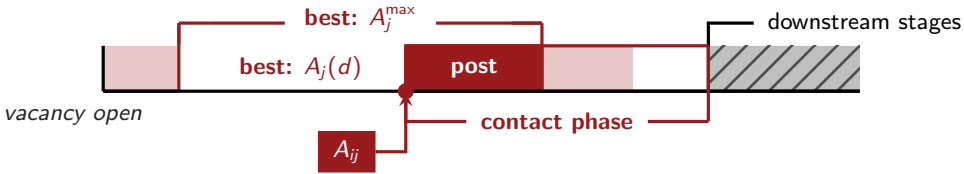


▸ fit summary ▸ robustness

The Screening-Value Cost of Early Contact Initiation



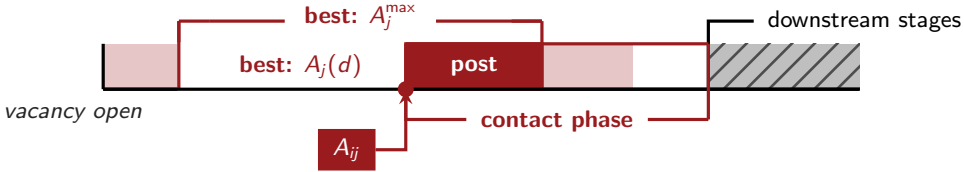
The Screening-Value Cost of Early Contact Initiation



If initiation is assigned to experimental application i, then

$$A_j^{max} - A_{ij} = \underbrace{A_j^{max} - A_j(d)}_{\text{timing component}} + \underbrace{A_j(d) - A_{ij}}_{\text{within-pool value gap}}$$

The Screening-Value Cost of Early Contact Initiation



If initiation is assigned to experimental application i, then

$$A_j^{\max} - A_{ij} = \underbrace{A_j^{\max} - A_j(d)}_{\text{timing component}} + \underbrace{A_j(d) - A_{ij}}_{\text{within-pool value gap}}$$

Construct two *conditional* loss measures:

$$\text{Missed-Best Probability} = \Pr\left(A_j(d) < A_j^{\max}\right), \quad \text{Relative Timing Loss} = \frac{\mathbb{E}\left[A_j^{\max} - A_j(d)\right]}{\mathbb{E}[A_{ij}]}$$

Early Contact Initiation Has Meaningful Screening-Value Costs

	$\bar{P}^{\exp}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	0.4079 [0.3985, 0.4177]	**0.5156** [0.5069, 0.5234]	**0.0656** [0.0642, 0.0674]

robustness

Early Contact Initiation Has Meaningful Screening-Value Costs

	$\bar{P}^{\text{exp}}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	0.4079 [0.3985, 0.4177]	**0.5156** [0.5069, 0.5234]	**0.0656** [0.0642, 0.0674]
One Day Earlier Δ		0.1175 [0.1144, 0.1209]	0.0217 [0.0211, 0.0223]
One Day Later Δ		-0.1713 [-0.1733, -0.1691]	-0.0292 [-0.0301, -0.0285]

▸ robustness

Early Contact Initiation Has Meaningful Screening-Value Costs

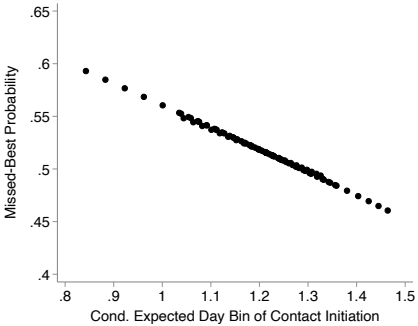
	$\bar{P}^{\text{exp}}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	0.4079 [0.3985, 0.4177]	**0.5156** [0.5069, 0.5234]	**0.0656** [0.0642, 0.0674]
One Day Earlier Δ		0.1175 [0.1144, 0.1209]	0.0217 [0.0211, 0.0223]
One Day Later Δ		-0.1713 [-0.1733, -0.1691]	-0.0292 [-0.0301, -0.0285]

$$\text{Wage-equivalent timing loss} = \underbrace{\text{RTL}}_{0.0656} \times \underbrace{\bar{w}}_{\$47{,}150} \times \underbrace{\Delta p_{\text{contact}}}_{63.12\%} \times \underbrace{\pi_{H|C}}_{[0.20,0.40]} = [\$390, \$781]$$

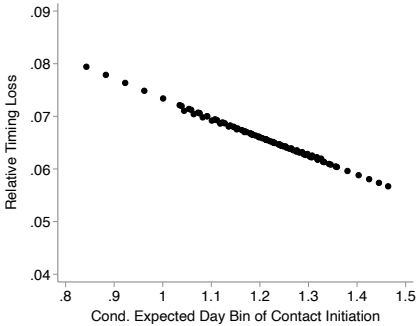
▸ robustness

Firms' Tradeoff: Speed-Loss Frontier

Across recruiting environments, earlier contact comes with larger timing losses:



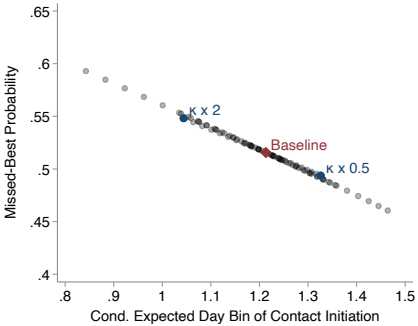
Missed-Best Probability



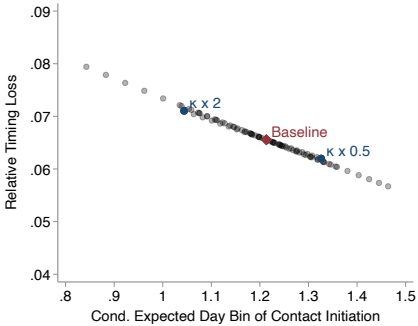
Relative Timing Loss

Firms' Tradeoff: Speed-Loss Frontier

Across recruiting environments, earlier contact comes with larger timing losses:



Missed-Best Probability

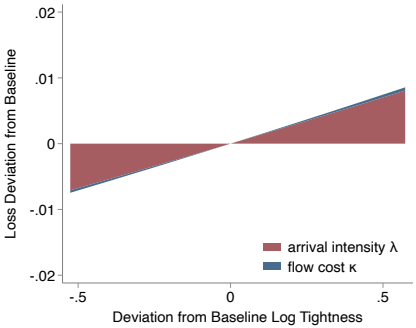


Relative Timing Loss

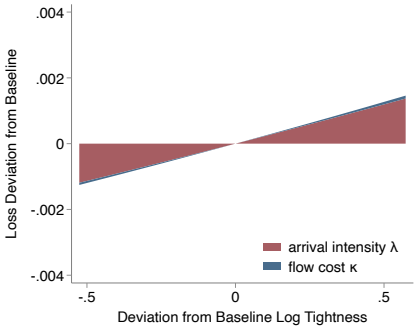
▸ magnitude

Quantifying Firms' Tradeoff: Counterfactuals

Tighter markets increase timing losses, mainly through the **arrival-intensity channel**:



Missed-Best Probability

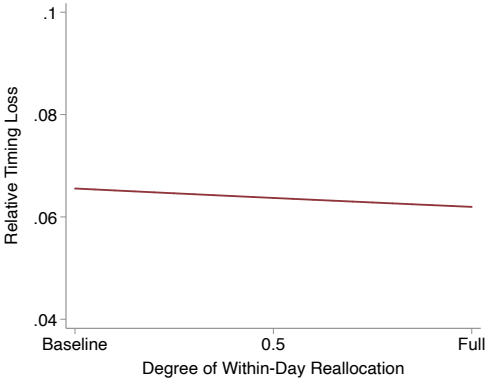


Relative Timing Loss

▸ other counterfactuals

Quantifying Firms' Tradeoff: Counterfactuals

Within-pool reallocation **cannot eliminate timing loss**:



Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

- **firm competition**: recruiting speed is a margin alongside wages and amenities (Carrillo-Tudela et al., 2023)

▸ external validity

Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

- **firm competition**: recruiting speed is a margin alongside wages and amenities (Carrillo-Tudela et al., 2023)
  ▸ external validity
- **speed premium**: applicants have an incentive to reallocate effort toward being early (Davis and de la Parra, 2024; Stanton and Thomas, 2025)

Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

- **firm competition**: recruiting speed is a margin alongside wages and amenities (Carrillo-Tudela et al., 2023)
  ▸ external validity
- **speed premium**: applicants have an incentive to reallocate effort toward being early (Davis and de la Parra, 2024; Stanton and Thomas, 2025)
  - **arms race to be early** (Horton et al., 2024)
  - **sorting by constraint** (Bandiera et al., 2025; Philippe and Skandalis, 2026; Tekleselassie et al., 2025)

Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

- **firm competition**: recruiting speed is a margin alongside wages and amenities (Carrillo-Tudela et al., 2023)
  ▸ external validity
- **speed premium**: applicants have an incentive to reallocate effort toward being early (Davis and de la Parra, 2024; Stanton and Thomas, 2025)
  - **arms race to be early** (Horton et al., 2024)
  - **sorting by constraint** (Bandiera et al., 2025; Philippe and Skandalis, 2026; Tekleselassie et al., 2025)
- **micro-to-macro link**: recruiting conditions shape when firms act and who receives consideration

Welfare Implications of Timing Frictions

Sequential arrival creates a queue-position wedge **even when firms optimize**:

- **firm competition**: recruiting speed is a margin alongside wages and amenities (Carrillo-Tudela et al., 2023)
  ▸ external validity
- **speed premium**: applicants have an incentive to reallocate effort toward being early (Davis and de la Parra, 2024; Stanton and Thomas, 2025)
  - **arms race to be early** (Horton et al., 2024)
  - **sorting by constraint** (Bandiera et al., 2025; Philippe and Skandalis, 2026; Tekleselassie et al., 2025)
- **micro-to-macro link**: recruiting conditions shape when firms act and who receives consideration
  - **time-varying matching efficiency** (Barlevy et al., 2024; Barnichon and Figura, 2015; Davis, Faberman, et al., 2006; Sedláček, 2014)
  - **cyclical match quality** (Barlevy, 2002; Bellou and Kaymak, 2021; Bils et al., 2023)

Conclusion

Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:

Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:
  - **Significance**: **1.2 p.p.** contact gap, *0.4 p.p. daily*, equivalent to 57% of the racial contact gap

Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:
  - **Significance**: **1.2 p.p.** contact gap, *0.4 p.p. daily*, equivalent to 57% of the racial contact gap
  - **Firm behavior**: **threshold-triggered contact initiation** is the key mechanism

Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:
  - **Significance**: **1.2 p.p.** contact gap, *0.4 p.p. daily*, equivalent to 57% of the racial contact gap
  - **Firm behavior**: **threshold-triggered contact initiation** is the key mechanism
  - **Cyclicality**: the timing gradient is **steeper** in tighter labor markets

Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:
  - **Significance**: **1.2 p.p.** contact gap, *0.4 p.p. daily*, equivalent to 57% of the racial contact gap
  - **Firm behavior**: **threshold-triggered contact initiation** is the key mechanism
  - **Cyclicality**: the timing gradient is **steeper** in tighter labor markets
- **Welfare implications** of the timing wedge
  - A within-vacancy mechanism of **application inflow** and **flow cost**
  - Quantifying the cost of the **timing friction**
  - Microfounding **cyclicality** of matching efficiency and quality

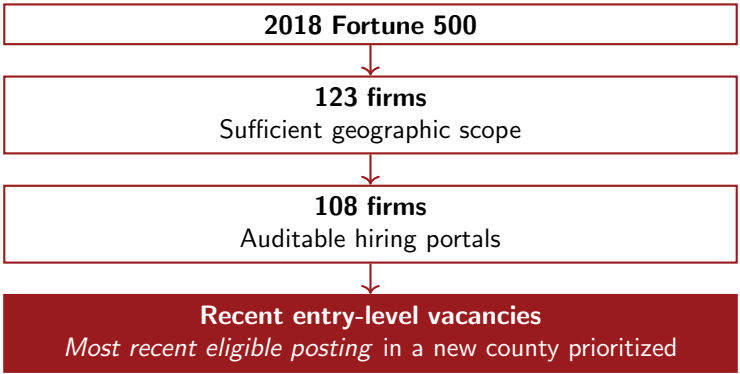
Timing Is Part of the Matching Process

- Novel evidence on **early application advantages** in job applications:
  - **Significance**: **1.2 p.p.** contact gap, *0.4 p.p. daily*, equivalent to 57% of the racial contact gap
  - **Firm behavior**: **threshold-triggered contact initiation** is the key mechanism
  - **Cyclicality**: the timing gradient is **steeper** in tighter labor markets
- **Welfare implications** of the timing wedge
  - A within-vacancy mechanism of **application inflow** and **flow cost**
  - Quantifying the cost of the **timing friction**
  - Microfounding **cyclicality** of matching efficiency and quality
- **Future work** can
  - embed the application-timing wedge in a **general equilibrium** model
  - study how recruiters **update their beliefs**

Thank you!

Contact me: saizhang@usc.edu

Sampling Firms and Vacancies

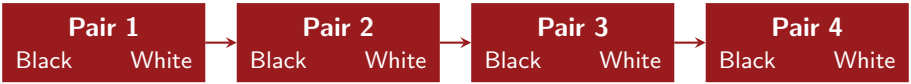


Target: 25 vacancies per firm-wave; 5 waves, 2019–2021.

back

Application Protocol and Randomization

- **submission protocol:** up to **8 applications** per vacancy in **4 pairs**



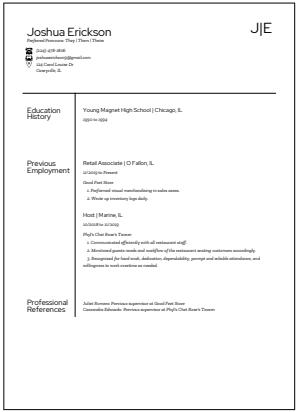
- **timing**: typically every other day in **Wave 1**, typically daily in **Wave 2–5**
- **other randomized attributes**:

Identity signals	**Credentials**	**Other attributes**
Gender, age, clubs, pronouns	Education, work experience	Resume design, personality-test responses

- **outcome:** employer contact within 30 days, phone numbers and/or email addresses recorded.

back

Example Experimental Resumes



Joshua Erickson

J|E

Perfected Pronouns: They | Them | Theirs

(555) 478-9846
joshuaerickson@gmail.com
144 Deer Lane Dr
Caterpillia, IL

Education History

Young Magnet High School | Chicago, IL
09/14 to 05/18

Previous Employment

Retail Associate | O Fallon, IL
07/2019 to Present

Good Feet Store

1. Performed visual merchandising to solve areas.
2. Wrote up inventory logs daily.

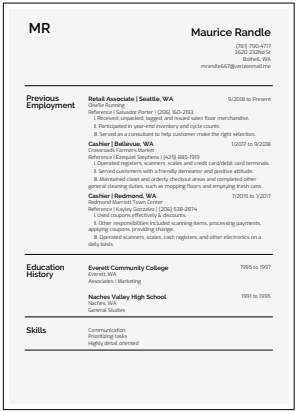
Host | Marine, IL
10/2018 to 12/2019

Pig's Chat Boat's Tavern

1. Communicated efficiently with all restaurant staff.
2. Monitored guests needs and workflow of the restaurant seating customers accordingly.
3. Recognized for hard work, dedication, dependability, prompt and reliable attendance, and willingness to work overtime as needed.

Professional References

Juliet Romeo Previous supervisor at Good Feet Store
Cassandra Edwards- Previous supervisor at Pig's Chat Boat's Tavern



MR

Maurice Randle

(787) 790-4717
3620 232Nd St
Bothell, WA
mrandle667@verizonmail.me

Previous Employment

Retail Associate | Seattle, WA 9/2018 to Present
Gisella Running
Reference | Salvador Porter | (206) 160-2793
I. Received, unpacked, tagged, and issued sales floor merchandise.
II. Participated in year-end inventory and cycle counts.
III. Served as a consultant to help customer make the right selection.

Cashier | Bellevue, WA 1/2017 to 9/2018
Crossroads Farmers Market
Reference | Ezequiel Stephens | (425) 885-1919
I. Operated registers, scanners, scales and credit card/debit card terminals.
II. Served customers with a friendly demeanor and positive attitude.
III. Maintained clean and orderly checkout areas and completed other general cleaning duties, such as mopping floors and emptying trash cans.

Cashier | Redmond, WA 7/2015 to 1/2017
Redmond Marriott Town Center
Reference | Kayley Gonzalez | (206) 538-2874
I. Used coupons effectively & discounts.
II. Other responsibilities included scanning items, processing payments, applying coupons, providing change.
III. Operated scanners, scales, cash registers, and other electronics on a daily basis.

Education History

Everett Community College 1995 to 1997
Everett, WA
Associates | Marketing

Naches Valley High School 1991 to 1995
Naches, WA
General Studies

Skills

Communication
Prioritizing tasks
Highly detail oriented

Source: Kline et al. (2022)

back

Market Conditions and Other Covariates

- **Main measure at experiment onset**: state-month labor-market tightness
  - job openings per unemployed worker (JOLTS and LAUS)
- **Additional recruiting conditions**
  - hires, quits, layoffs and discharges, and job-opening rates (JOLTS)
- **Descriptive external-validity covariates**
  - industry and job content (O*NET; BLS CES and ECI)
  - state income, institutions, and employment composition (IDDA, Kondo et al., 2023) (DOL; LAUS)
  - firm organization, financial characteristics, and job postings (Compustat; Revelio Labs)

back

Construction of Application Timing Measures

	$early_i$	$early_i^2$	$early_i^3$	$Order_i$	$D2L_i^t$	$N(Contacts)_i$
$early_i$	1					
$early_i^2$	0.9786	1				
$early_i^3$	0.9569	0.9710	1			
$Order_i$	-0.8164	-0.8076	-0.8006	1		
$D2L_i^t$	0.7169	0.7195	0.7146	-0.7406	1	
$N(Contacts)_i$	-0.2048	-0.2006	-0.1993	0.2385	-0.1790	1

- $early_i = \mathbf{1}$ (application i among the first 4 applications)
- $early_i^2 = \mathbf{1}$ (application i not among the last 4 applications)
- $early_i^3 = \mathbf{1}$ (application i not among the last 2 pairs of applications)

back

Balance of Timing Measures

Dep.Var.	(1) $early^1$	(2) $early^2$	(3) $early^3$	(4) $Order$	(5) $Order^{pair}$	(6) $D2L^t$
Balanced	1.55	1.37	1.19	0.73	1.02	1.25
	[0.1228]	[0.1938]	[0.2967]	[0.6864]	[0.4212]	[0.2568]
Full	1.07	0.98	0.83	0.63	0.83	0.97
	[0.3831]	[0.4564]	[0.5857]	[0.7692]	[0.5876]	[0.4618]
Complete	0.99	0.99	0.82	0.53	1.11	0.92
	[0.4481]	[0.4481]	[0.5988]	[0.8540]	[0.3519]	[0.5040]
Incomplete	1.10	1.06	0.87	0.93	1.03	0.75
	[0.3566]	[0.3931]	[0.5478]	[0.5007]	[0.4142]	[0.6610]

back

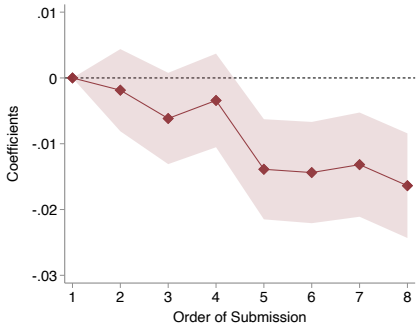
Robustness: Early Indicator

	(1) Complete	(2) Incomplete	(3) Wave 1	(4) Waves 2–5	(5) Full Sample	(6) Full Sample
$early_i$	0.0112^{***}	0.0240^{*}	0.0128^{**}	0.0113^{***}		
	(0.0024)	(0.0127)	(0.0061)	(0.0025)		
$early_i^2$					0.0135^{***}	
					(0.0023)	
$early_i^3$						0.0105^{***}
						(0.0023)
N	78,632	5,011	14,541	69,102	83,643	83,643

▸ construction ▸ back

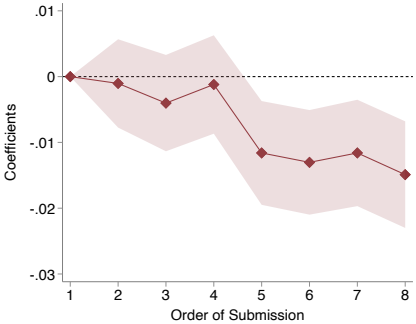
Early Application Advantage: Submission Order

$$Y_i = \alpha + \sum_{k=2}^{8} \beta_k \mathbf{1}\{Order_i = k\} + \boldsymbol{\gamma}' \mathbf{X}_i + \eta_{j(i)} + \varepsilon_i$$

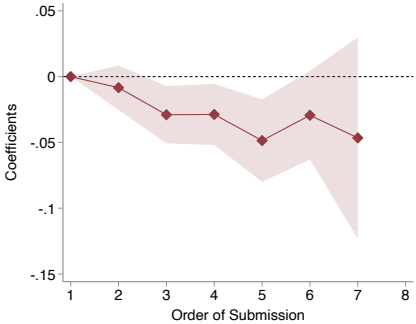


robustness back

Robustness: Submission Order



Complete vacancies



Incomplete vacancies

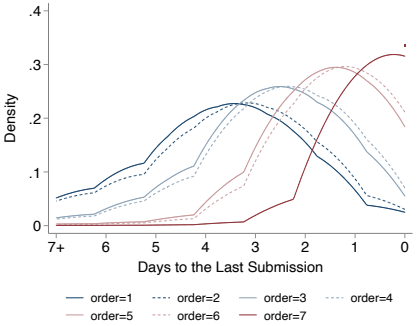
back

Robustness: Days to Last Submission

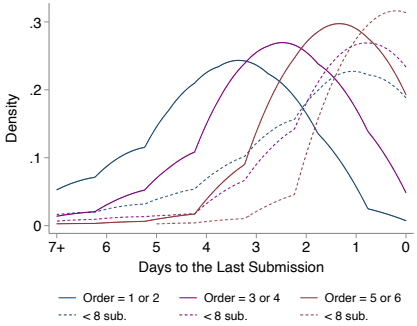
	(1) Full Sample	(2) Wave 1	(3) Waves 2–5	(4) Complete	(5) Incomplete
$D2L_i$ (w.)	0.0039^{***}	0.0034^{***}	0.0042^{***}		
	(0.0008)	(0.0013)	(0.0010)		
$D2L_i^t$				0.0038^{***}	0.0087^{**}
				(0.0008)	(0.0035)
N	83,643	14,541	69,102	78,632	5,011

back

Robustness: Days to Last Submission



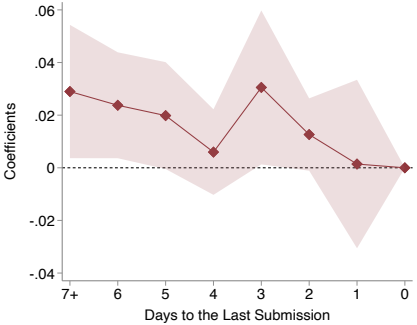
Full Sample



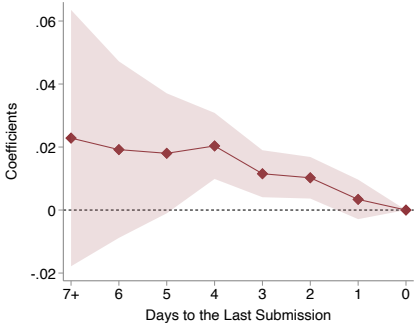
Complete vs. Incomplete

back

Robustness: Days to Last Submission



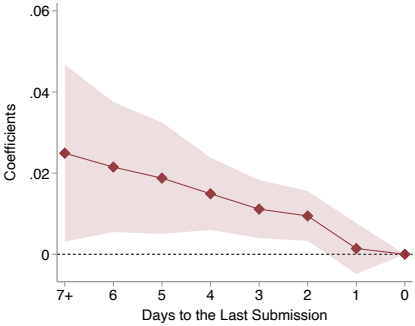
Wave 1



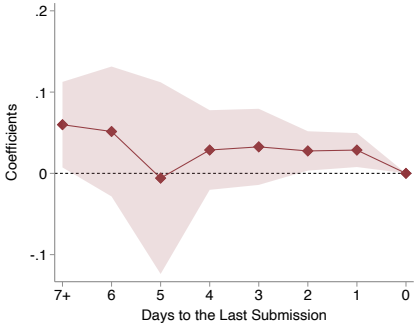
Waves 2–5

back

Robustness: Days to Last Submission



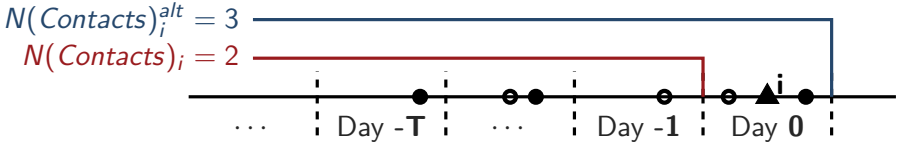
Complete



Incomplete

back

Constructing the Post-Initiation Indicator



	$N(Contacts)_i^{alt} = 0$	$N(Contacts)_i^{alt} > 0$	Diff.
	.2686***	.1536***	**-.1150***
	(.0035)	(.0116)	(.0123)
N	67,082	11,550	78,632

back

Contact Initiation Absorbs the Timing Advantage

	(1)	(2)	(3)	(4)
$early_i$	0.0112***	-0.0017		
	(0.0024)	(0.0022)		
$D2L_i^t$			0.0038***	-0.0011
			(0.0008)	(0.0008)
$\mathbf{1}(N(Contacts)_i^{alt} > 0)$		-0.1257***		-0.1273***
		(0.0094)		(0.0094)
N		78,632		

back

Machine Learning Prediction Models

- **Training sample**: first experimental application to each vacancy, $N = 11,114$
- **Input predictors**: all resume characteristics, SOC3 categories of vacancies, detailed resume entry categories (1,512 job-type categories, 23 job titles, duration in months)
- **Out-of-fold construction**:
  - Posting-level five-fold cross-fitting
  - Each vacancy scored using a model trained on other vacancies
- **Model candidates**: Elastic-net logistic regression, ExtraTrees classifier, CatBoost gradient boosting (with native and encoded category handling)
- **Model selection criteria**: prediction accuracy and within-vacancy score dispersion

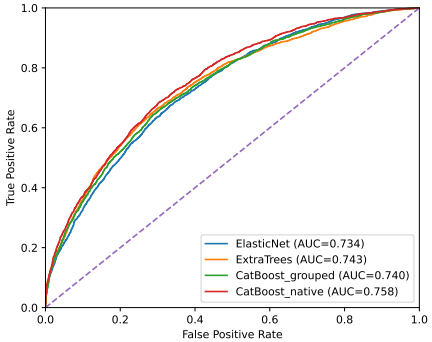
▸ performance score ▸ visual comparison ▸ back

Pre-Tuning Model Comparison

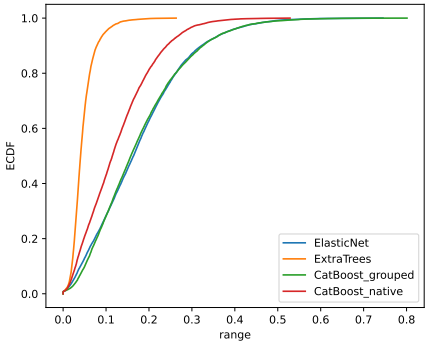
Model	ROC-AUC	Log Loss	Brier	Average Precision
CatBoost (native categories)	0.758	0.474	0.154	0.525
ExtraTrees (encoded)	0.743	0.491	0.160	0.511
CatBoost (encoded)	0.740	0.484	0.158	0.504
Elastic-net logit (encoded)	0.734	0.491	0.161	0.488

▸ model details ▸ back

Machine Learning Model Comparison



ROC Curve



Predicted Score Range

▸ model details ▸ back

Prediction Score Validation

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies			Vacancies w. ≥ 1 Contact		
Main index	0.1587***			0.2807***		
	(0.0404)			(0.0715)		
Alternative 1		0.1240***			0.2525***	
		(0.0282)			(0.0573)	
Alternative 2			0.1078***			0.2060***
			(0.0278)			(0.0533)
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	32,288	32,288	32,288

back

Prediction Score Validation

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies			Vacancies w. ≥ 1 Contact		
Main index	1.0120***			0.6084***		
	(0.0182)			(0.0253)		
Alternative 1		1.1650***			0.6988***	
		(0.0214)			(0.0293)	
Alternative 2			1.1340***			0.6743***
			(0.0208)			(0.0285)
St-Mth FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	32,288	32,288	32,288

back

Prediction Score Validation

Dep.Var.	(1)	(2)	(3)	(4)	(5)	(6)
	Main Index		Alternative 1		Alternative 2	
$early_i^1$	-0.0003		0.0004		0.0003	
	(0.0002)		(0.0003)		(0.0003)	
$D2L_i^t$		-0.0001		0.0002		0.0001
		(0.0001)		(0.0001)		(0.0001)
Vacancy FEs	✓	✓	✓	✓	✓	✓
N	78,632	78,632	78,632	78,632	78,632	78,632

back

IV Estimation: Robustness

Using an alternative index built from a CatBoost model with frequency-based grouping

	All Vacancies		Contacted Vacancies	
	(1) IV-2SLS	(2) OLS	(3) IV-2SLS	(4) OLS
$\mathbf{1}(N(Contacts)_i > 0)$	-0.6520	-0.1205***	-0.5742*	-0.1277***
	(0.3981)	(0.0206)	(0.3469)	(0.0217)
Kleibergen-Paap F	10.781		15.172	
Anderson-Rubin p	0.083		0.085	
N	39,300		16,142 (4,036 *vacancies*)	

back

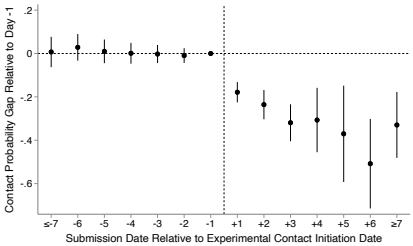
IV Estimation: Robustness

Using an alternative index built from a CatBoost model with text-based grouping

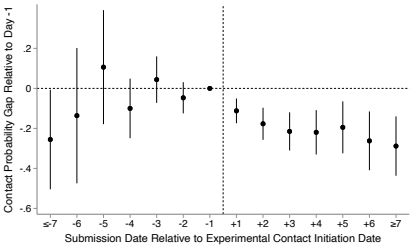
	All Vacancies		Contacted Vacancies	
	(1) IV-2SLS	(2) OLS	(3) IV-2SLS	(4) OLS
$\mathbf{1}(N(Contacts)_i > 0)$	-1.0857	-0.1204***	-0.9264	-0.1275***
	(0.8640)	(0.0206)	(0.6883)	(0.0217)
Kleibergen-Paap F	3.406		5.112	
Anderson-Rubin p	0.112		0.113	
N	39,300		16,142 (4,036 *vacancies*)	

back

Contact Initiation and the Best Application Arrival



Before Initiation



After Initiation

▸ main event study ▸ back

Alternative Explanation: Pool-Quality Learning

	(1)	(2)	(3)	(4)
	Outcome: Contact Initiation			
	All Vacancies		Vacancies w. ≥ 1 Contact	
Q_i^{best}	0.0078**		0.0235**	
	(0.0039)		(0.0098)	
Q_i^{mean}		0.0040		0.0134
		(0.0079)		(0.0180)
N	39,300	39,300	16,142	16,142

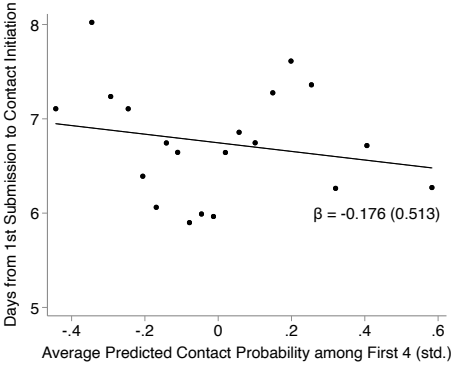
back

Alternative Explanation: Pool-Quality Learning

	(1)	(2)	(3)	(4)	(5)	(6)
	Outcome: Contact					
	All Vacancies			Vacancies w. ≥ 1 Contact		
Q_i^{best}	-0.0102**		-0.0099	-0.0271**		-0.0274*
	(0.0050)		(0.0064)	(0.0130)		(0.0165)
Q_i^{mean}		-0.0129	-0.0007		-0.0309	0.0008
		(0.0098)	(0.0126)		(0.0233)	(0.0295)
N	39,300	39,300	39,300	16,142	16,142	16,142

back

Alternative Explanation: Pool-Quality Learning



back

Alternative Explanation: Intrinsic Value of Earliness

	(1)	(2)	(3)	(4)	(5)	(6)
	All Vacancies		Early Contact Init.		Late Contact Init.	
early	0.0273***		0.0933***		-0.0523***	
	(0.0058)		(0.0076)		(0.0085)	
$D2L^t$		0.0096***		0.0326***		-0.0243***
		(0.0020)		(0.0025)		(0.0031)
N	32,288	32,288	17,616	17,616	14,672	14,672

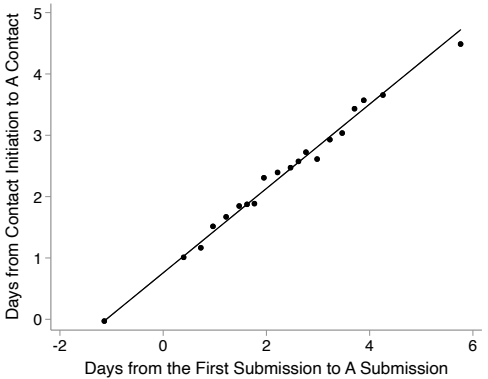
back

Alternative Explanation: Full-Pool Comparison

	(1)	(2)	(3)	(4)
	All Vacancies		Vacancies w. $\geq$ 1 Contact	
$Q_i^{best,post}$	0.0053		0.0122	
	(0.0048)		(0.0118)	
$Q_i^{mean,post}$		0.0027		0.0044
		(0.0099)		(0.0233)
N	39,281	39,281	16,143	16,143

back

Alternative Explanation: Pure Batching



back

Contact Timing by Labor Market Tightness: Robustness

	(1)	(2)	(3)	(4)	(5)
			Pr (Contact Initiator)		
$D2F_i^t \times \log(\text{hires}_{m(i),t(i)})$	-0.0097*				-0.0054
	(0.0052)				(0.0063)
$D2F_i^t \times \log(\text{layoffs}_{m(i),t(i)})$		0.0006			-0.0062***
		(0.0011)			(0.0014)
$D2F_i^t \times \log(\text{quits}_{m(i),t(i)})$			-0.0166***		0.0021
			(0.0047)		(0.0053)
$D2F_i^t \times \log(\text{job opening}_{m(i),t(i)})$				-0.0278***	0.0024
				(0.0055)	(0.0103)
$D2F_i^t \times \log(\text{tightness}_{m(i),t(i)})$					-0.0368***
					(0.0080)
N	78,632	78,632	78,632	78,632	78,632

back

Construction of $D2F_i^t$

$D2F_i^t$ is constructed similarly to $D2L_i^t$:

$$D2F_i^t = \min\{\text{days between application } i \text{ and the first application}, 7\}$$

$D2F_i^t$ and $D2L_i^t$ are almost perfectly negatively correlated:

	$D2F_i^t$
$D2L_i^t$	-0.9748***
	(0.0018)
Vacancy FEs	✓
N	78,632

back

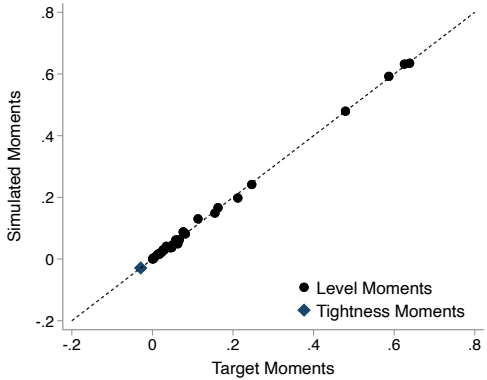
Targeted Moments

Level moments: Contact-initiation day mass

	$d=0$	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$	$d=6$	None
$h(j)=1$	0.1628	0.0478	0.0345	0.0294	0.0263	0.0179	0.0234	0.6384
	(0.0088)	(0.0051)	(0.0042)	(0.0038)	(0.0036)	(0.0030)	(0.0034)	(0.0118)
$h(j)=2$	0.2120	0.0768	0.0580	0.0446	0.0123	0.0035	0.0026	0.5867
	(0.0088)	(0.0050)	(0.0041)	(0.0037)	(0.0022)	(0.0011)	(0.0011)	(0.0111)
$h(j)=3$	0.1552	0.0766	0.0670	0.0503	0.0174	0.0039	0.0021	0.6255
	(0.0051)	(0.0035)	(0.0032)	(0.0029)	(0.0018)	(0.0009)	(0.0006)	(0.0072)
$h(j)=4$	0.2465	0.1131	0.0818	0.0629	0.0134	0.0018	0.0005	0.4792
	(0.0083)	(0.0060)	(0.0048)	(0.0045)	(0.0021)	(0.0009)	(0.0005)	(0.0107)
Tightness moment							-0.0291	
							(0.0037)	

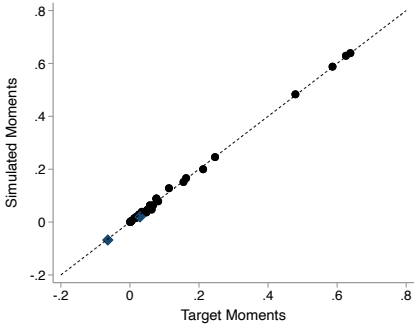
back

Moment Fitting

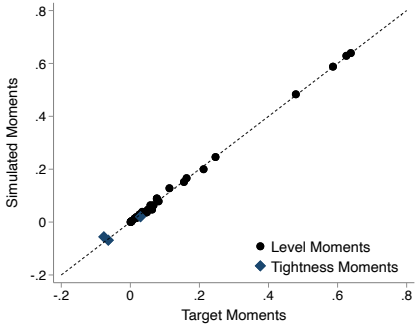


back

Moment Fitting Robustness



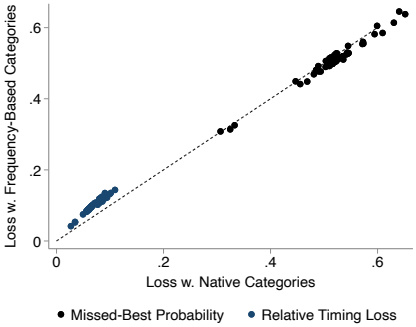
Alternative Moment Set A



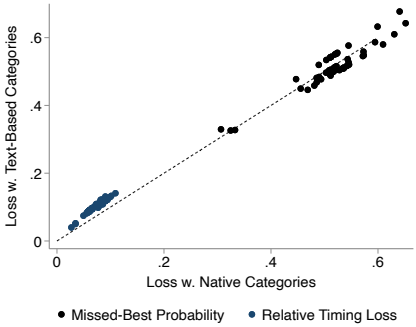
Alternative Moment Set B

back

Headline Result Robustness: Screening-Value Costs



Frequency-Based Category Scores



Text-Based Category Scores

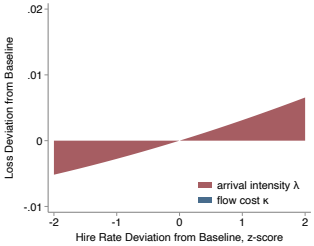
back

Selected Counterfactuals on the Frontier

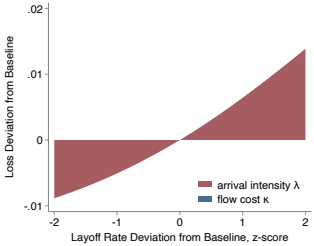
Counterfactual	Expected Initiation-Day Bin	$\bar{P}^{\exp}(\widehat{\Theta})$	Missed-Best Probability	Relative Timing Loss
Baseline	1.213	0.4079	0.5156	0.0656
$\kappa_j \times 0.5$	1.326	0.3726	0.4937	0.0620
$\kappa_j \times 2$	1.044	0.4579	0.5482	0.0711
Mean $\log(\theta)$ to p_{10}	1.248	0.3754	0.5081	0.0643
Mean $\log(\theta)$ to p_{90}	1.182	0.4359	0.5221	0.0667

back

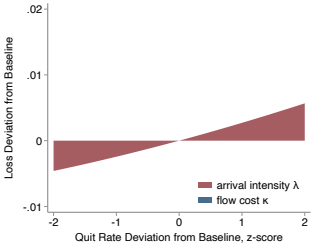
Quantifying Firms' Tradeoff: Counterfactuals



by Hires



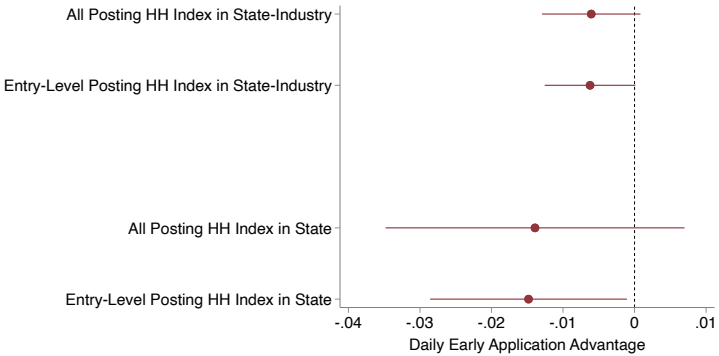
by Layoffs/Discharges



by Quits

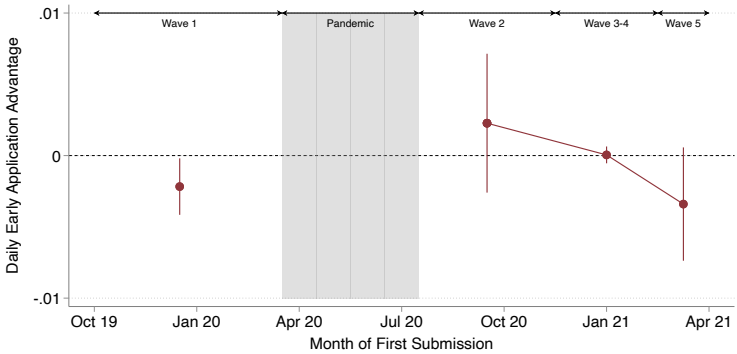
back

External Validity: Market Concentration



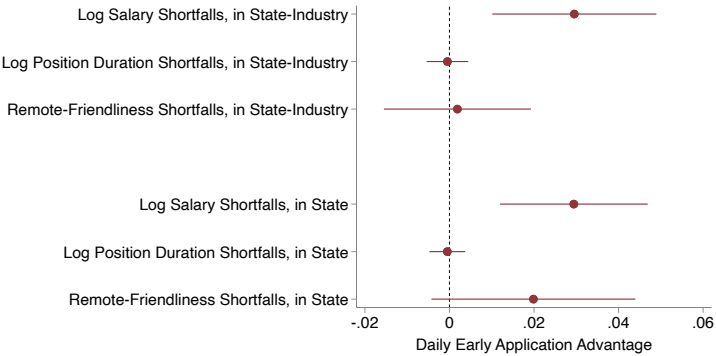
▸ welfare implications

External Validity: Local Recruitment Prominence



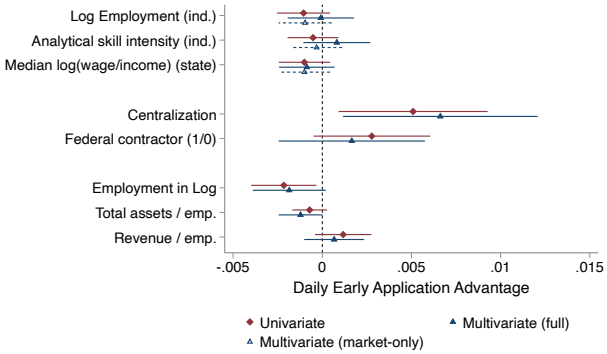
▸ welfare implications

External Validity: Offer Competitiveness



▸ welfare implications

External Validity: Market, Firm and Vacancy Characteristics



▸ early application advantage ▸ welfare implications

References I

Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, *138*(1), 1–35.

Bandiera, O., Jalal, A., & Roussille, N. (2025). *The illusion of time: Gender gaps in job search and employment*. (Working Paper No. 34051). National Bureau of Economic Research. doi:10.3386/w34051

Barlevy, G. (2002). The sullying effect of recessions. *The Review of Economic Studies*, *69*(1), 65–96.

Barlevy, G., Faberman, R. J., Hobijn, B., & Şahin, A. (2024). The shifting reasons for beveridge curve shifts. *Journal of Economic Perspectives*, *38*(2), 83–106.

Barnichon, R., & Figura, A. (2015). Labor market heterogeneity and the aggregate matching function. *American Economic Journal: Macroeconomics*, *7*(4), 222–249.

Bartoš, V., Bauer, M., Chytilová, J., & Matějka, F. (2016). Attention discrimination: Theory and field experiments with monitoring information acquisition. *American Economic Review*, *106*(6), 1437–1475.

References II

Bellou, A., & Kaymak, B. (2021). The cyclical behavior of job quality and real wage growth. *American Economic Review: Insights*, *3*(1), 83–96.

Bils, M., Kudlyak, M., & Lins, P. (2023). The quality-adjusted cyclical price of labor. *Journal of Labor Economics*, *41*(S1), S13–S59.

Birinci, S., See, K., & Wee, S. L. (2025). Job applications and labour market flows. *Review of Economic Studies*, *92*(3), 1438–1496.

Carrillo-Tudela, C., Gartner, H., & Kaas, L. (2023). Recruitment policies, job-filling rates, and matching efficiency. *Journal of the European Economic Association*, *21*(6), 2413–2459.

Cortés, P., Pan, J., Pilossoph, L., Reuben, E., & Zafar, B. (2023). Gender differences in job search and the earnings gap: Evidence from the field and lab. *The Quarterly Journal of Economics*, *138*(4), 2069–2126.

Davis, S. J., & de la Parra, B. S. (2024). *Application flows*. National Bureau of Economic Research.

Davis, S. J., Faberman, R. J., & Haltiwanger, J. (2006). The flow approach to labor markets: New data sources and micro–macro links. *Journal of Economic perspectives*, *20*(3), 3–26.

References III

- Gaddis, S. M. (Ed.). (2018). *Audit Studies: Behind the Scenes with Theory, Method, and Nuance*. doi:10.1007/978-3-319-71153-9
- Hoffman, M., Kahn, L. B., & Li, D. (2018). Discretion in hiring. *The Quarterly Journal of Economics*, *133*(2), 765–800.
- Horton, J. J., Vasserman, N., Stanford, G., & Watt, N. (2024). *Reducing congestion in labor markets: A case study in simple market design*. Technical Report, National Bureau of Economic Research.
- Kessler, J. B., Low, C., & Sullivan, C. D. (2019). Incentivized resume rating: Eliciting employer preferences without deception. *American Economic Review*, *109*(11), 3713–3744. doi:10.1257/aer.20181714
- Kline, P., Rose, E. K., & Walters, C. R. (2022). Systemic discrimination among large u.s. employers. *The Quarterly Journal of Economics*, *137*(4), 1963–2036.

References IV

Kondo, I., Rinz, K., Hawkins, B., Gubbay, N., Voorheis, J., & Wozniak, A. (2023). *Granular income inequality and mobility using idda*. (Working Paper No. CES-23-55). U.S. Census Bureau Center for Economic Studies. Associated dataset: version 1.0. Retrieved from https://doi.org/10.21034/data.idda

Neumark, D. (2018). Experimental research on labor market discrimination. *Journal of Economic Literature*, *56*(3), 799–866. doi:10.1257/jel.20161309

Philippe, A., & Skandalis, D. (2026). *Motherhood and constraints during job search*. (Discussion Paper No. 21431). Centre for Economic Policy Research. Paris & London. Retrieved from https://cepr.org/publications/dp21431

Radbruch, J., & Schiprowski, A. (2025). Interview sequences and the formation of subjective assessments. *Review of Economic Studies*, *92*(2), 1226–1256.

Sedláček, P. (2014). Match efficiency and firms' hiring standards. *Journal of Monetary Economics*, *62*, 123–133.

Stanton, C. T., & Thomas, C. (2025). Who benefits from online gig economy platforms? *American Economic Review*, *115*(6), 1857–1895.

References V

Tekleselassie, T., Witte, M., Radbruch, J., Hensel, L., & Isphording, I. E. (2025). *Feedback, confidence and job search behavior*. IZA Institute of Labor Economics. IZA Discussion Paper.

Wolthoff, R. (2018). Applications and interviews: Firms' recruiting decisions in a frictional labour market. *The Review of Economic Studies*, *85*(2), 1314–1351.

Wright, R., Kircher, P., Julien, B., & Guerrieri, V. (2021). Directed search and competitive search equilibrium: A guided tour. *Journal of Economic Literature*, *59*(1), 90–148.